Experiment No: 4

Course Outcome: CO4 **Blooms Level: L4**

Aim: Sentiment Analysis and visualization of Social Media Text Data for business.

Abstract:

This experiment focuses on Exploratory Data Analysis. It performs Sentiment Analysis. Start with data collection, preprocessed that data and generate word cloud visualization for positive, negative and neutral sentiments. This process helps understand public opinion, analyze customer feedback, improve marketing strategies.

Sample Input and Output:

Case 1: Provide the YouTube video ID and API key to the YouTube Data API.
Output: YouTube comments in structured format (CSV/JSON) are successfully saved for further analysis.

Case 2: Preprocess the data.
Output: Data Cleaning, filtration and Storage involved in preprocess method.

Case 3: Generate word count Visualization
Output: Detect positive, negative, or neutral sentiment

Theory:

- TextBlob: Python library for processing textual data. It provides a simple API for diving into common natural language processing (NLP) tasks. TextBlob outputs sentiment in terms of: Polarity, which quickly identify the overall mood of a text and Subjectivity, based on opinions, feelings, or personal views and helps to filter out irrelevant or biased texts, and focus on the genuine sentiments.

- Sentiment Analysis: is also known as opinion mining. It is a natural language processing technique used to identify, extract, and classify emotions or opinions expressed in **positive, negative, or neutral** form of given text data.

- Word Cloud: It is a powerful and straightforward visualization method, represents the most frequent words in a dataset by size. It Summarizes large text data visually and highlight the most important terms.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import nltk
nltk.download('stopwords')
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
from nltk.stem.porter import PorterStemmer

import string
import re
import textblob
from textblob import TextBlob
```

```
import os

from wordcloud import WordCloud, STOPWORDS
from wordcloud import ImageColorGenerator
import warnings
%matplotlib inline
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

```
#Read the JSON generated from the CLI command above and create a pandas dataframe
tweets_df = pd.read_csv(r'/content/sample_data/Tweets.csv')
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Tweets.csv to Tweets.csv
```

```
tweets_df.head(5)
```

	tweet_id	airline_sentiment	airline_sentiment_confidence	negativereason	negativereason_conf:
0	570306133677760513	neutral	1.0000	NaN	

```
tweets_df.to_csv()
```

```
',tweet_id,airline_sentiment,airline_sentiment_confidence,negativereason,negativereason_confidence,airline,
airline_sentiment_gold,name,negativereason_gold,retweet_count,text,tweet_coord,tweet_created,tweet_locatio
n,user_timezone\n0,570306133677760513,neutral,1.0,,,Virgin America,,cairdin,,0,@VirginAmerica What @dhepbur
n said.,,2015-02-24 11:35:52 -0800,,Eastern Time (US & Canada)\n1,570301130888122368,positive,0.3486,,0.0,V
irgin America,,jnardino,,0,@VirginAmerica plus you\'ve added commercials to the experience... tacky.,,2015-
02-24 11:15:59 -0800,,Pacific Time (US & Canada)\n2,570301083672813571,neutral,0.6837,,,Virgin America,,yvo
nnalynn,,0,@VirginAmerica I didn\'t today... Must mean I need to take another trip!,,2015-02-24 11:15:48 -0
800,Lets Play,Central Time (US & Canada)\n3,570301031407624196,negative,1.0,Bad Flight,0.7033,Virgin Americ
a,,jnardino,,0,"@VirginAmerica it\'s really aggressive to blast obnoxious ""entertainment"" in your guests
\' faces &amp; they have little recourse...'
```

```
tweets_df.shape
```

```
(14640, 15)
```

```
tweets_df.head
```

pandas.core.generic.NDFrame.head

```
def head(n: int=5) -> Self
```

Return the first `n` rows.

This function returns the first `n` rows for the object based on position. It is useful for quickly testing if your object has the right type of data in it.

```
import plotly.graph_objects as go
Top_Location_Of_tweet= tweets_df['airline'].value_counts().head (10)
```

```
print(Top_Location_Of_tweet)
```

```
airline
United            3822
US Airways        2913
American          2759
Southwest         2420
Delta             2222
Virgin America     504
Name: count, dtype: int64
```

```
from nltk. corpus import stopwords
stop = stopwords.words('english')
tweets_df['text'].apply(lambda x: [item for item in x if item not in stop])
tweets_df.shape
```

```
(14640, 15)
```

```
!pip install tweet-preprocessor
```

```
Requirement already satisfied: tweet-preprocessor in c:\users\kadambari deherkar\anaconda3\lib\site-pack
```

```
#Remove unnecessary characters
punct  =  ['%','/',':','\\','&amp','&',';','?']
```

```
tweets_df['text'] = tweets_df['text'].apply(lambda x: remove_punctuations(x))
```

```
#Drop tweets that has empty text fields
tweets_df['text'].replace( '', np.nan, inplace=True)
tweets_df.dropna(subset=["text"],inplace=True)
len(tweets_df)
```

```
14640
```

```
tweets_df = tweets_df.reset_index(drop=True)
tweets_df.head()
```

```
plt.show()


count_vectorizer = CountVectorizer(stop_words=stop)
count_data = count_vectorizer.fit_transform(tweets_df['text'])

# Visualize the 20 most common words
plot_20_most_common_words(count_data, count_vectorizer)
```

```
/usr/local/lib/python3.11/dist-packages/sklearn/feature_extraction/text.py:402: UserWarning: Your stop_
  warnings.warn(
<ipython-input-24-71514098d9d7>:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x`
```

```
import cufflinks as cf
cf.go_offline()
cf.set_config_file(offline=False, world_readable=True)

def get_top_n_bigram(corpus, n=None) :
  vec = CountVectorizer(ngram_range=(2, 4), stop_words="english").fit(corpus)
  bag_of_words = vec.transform(corpus)
  sum_words = bag_of_words.sum(axis=0)
  words_freq =[(word, sum_words[0, idx]) for word, idx in vec.vocabulary_.items()]
  words_freq =sorted(words_freq, key = lambda x: x[1], reverse=True)
  return words_freq[:n]

common_words = get_top_n_bigram(tweets_df['text'] , 8)
mydict={}
for word, freq in common_words:
  bigram_df = pd.DataFrame(common_words,columns = ['ngram', 'count'])

bigram_df.groupby( 'ngram' ).sum()['count'].sort_values(ascending=False).sort_values().plot.barh
```

```
<Axes: title={'center': 'Top 8 bigrams'}, ylabel='ngram'>
```

TextBlob

```
def get_subjectivity(text):
  return TextBlob(text).sentiment.subjectivity

def get_polarity(text):
  return TextBlob(text).sentiment.polarity
```

```
tweets_df['subjectivity']=tweets_df[ 'text'].apply(get_subjectivity)
tweets_df['polarity']=tweets_df[ 'text'].apply(get_polarity)
tweets_df.loc[:,['airline','text','airline_sentiment','subjectivity','polarity']].head(20)
```

** Sentiment Analysis**

```
tweets_df['textblob_score'] =tweets_df['text'].apply(lambda x: TextBlob(x).sentiment.polarity)
```

```
neutral_threshold=0.05
```

```
tweets_df['textblob_sentiment']=tweets_df[ 'textblob_score'].apply(lambda c:'positive' if c >= neutral_threshold else ('Negative'
```

```
textblob_df =  tweets_df[['text','textblob_sentiment','retweet_count']]
textblob_df
```

	text	textblob_sentiment	retweet_count
0	@VirginAmerica What @dhepburn said.	Neutral	0
1	@VirginAmerica plus you've added commercials t...	Neutral	0
2	@VirginAmerica I didn't today... Must mean I n...	Negative	0
3	@VirginAmerica it's really aggressive to blast...	Neutral	0
4	@VirginAmerica and it's a really big bad thing...	Negative	0
...	...	...	...
14635	@AmericanAir thank you we got on a different f...	Neutral	0
14636	@AmericanAir leaving over 20 minutes Late Flig...	Negative	0
14637	@AmericanAir Please bring American Airlines to...	Neutral	0

```
textblob_df["textblob_sentiment"].value_counts()
```

textblob_sentiment	count
Neutral	6091
positive	5210
Negative	3339

dtype: int64

```
df_positive=textblob_df[textblob_df['textblob_sentiment']=='positive' ]
```

+ Code + Text

```
df_very_positive=df_positive[df_positive['retweet_count']>0]
```

```
df_very_positive.head()
```

	text	textblob_sentiment	retweet_count
147	Always have it together!!! You're welcome! RT ...	positive	2
264	@VirginAmerica thanks for gate checking my bag...	positive	1
318	@VirginAmerica can u help this 🤦‍♀️ @FreyaBevan_F...	positive	2
336	@VirginAmerica Many Thanks for the Follow.\n#o...	positive	1
402	@VirginAmerica announced New Route to Austin w...	positive	3

```
df_negative=textblob_df[textblob_df['textblob_sentiment']=='Negative' ]
```

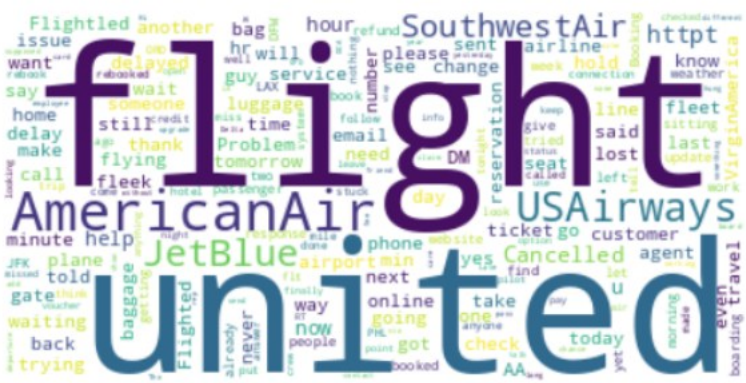
```
df_negative
```

Create a Word Cloud

```
from wordcloud import WordCloud, STOPWORDS
from PIL import Image
```

```
#Creating the text variable
positive_tw =" ".join(t for t in df_very_positive.text)
# Creating word _ cloud with text as argument in . generate() rtpthod
word_cloud1 = WordCloud(collocations = False, background_color = 'white') .generate(positive_tw)
# Display the generated Word Cloud
plt. imshow(word_cloud1, interpolation='bilinear')
plt.axis('off')
plt.show()
```

```
#Creating the text variable
neutral_tw =" ".join(t for t in df_neutral.text)
# Creating word _ cloud with text as argument in . generate() rtpthod
word_cloud2 = WordCloud(collocations = False, background_color = 'white') .generate(neutral_tw)
# Display the generated Word Cloud
plt. imshow(word_cloud2, interpolation='bilinear')
plt.axis('off')
plt.show()
```



Conclusion:

In this experiment, data was successfully collected from social media platforms. preprocessed that data and generate word cloud visualization for positive, negative and neutral sentiments. This process helps understand public opinion, analyze customer feedback, improve marketing strategies. Thus sentiment analysis and visualization of social media text data for business is performed.

Exercise 1:

Perform Sentiment Analysis on the data collected from YouTube using the YouTube Data API, store the extracted data in a structured .csv or JSON format, preprocessed this data. Generate word cloud visualization for positive, negative and neutral sentiments.

Abstract

This exercise focuses on extracting publicly available user reviews from Google. The collected data is stored in a structured format for ease of processing and analysis. The extracted reviews were preprocessed and a sentimental analysis is performed of sentiment in positive, negative and neutral format.

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Class: BE **Subject: Social Media Analytics** **Sem-VIII**

Exercise 2:

To perform comparative sentiment analysis on social media text data in order to compare public opinion and customer sentiment toward two or more competing brands

Abstract:

This exercise focuses on extracting publicly available data from social media platforms. Businesses need to understand how their brand is perceived in comparison to competitors on social media platforms. Show the visualization in bar diagram form. This exercise analyzes and compares sentiment polarity across multiple brands to support strategic business decisions.